Twelve promise-class conjectures for certified polynomial-time many-electron approximation

A companion collection to *Three hundred nineteen conjectures in mathematics*

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Abstract

This is a companion collection to the main conjecture suite, and it is held to a deliberately weaker and clearly separated standard of evidence. The main collection states exact, unconditional mathematical laws, each reproduced numerically to the stated precision. The twelve conjectures here are of a different genre: each is a *conditional, promise-class* statement asserting that, on an explicitly defined structural promise class, a certified bounded-error classical algorithm for a finite-basis many-electron ground-state problem runs in polynomial time. None of them claims an unconditional polynomial-time solver for arbitrary electronic Hamiltonians, which would conflict with the fermionic-sign, QMA-hardness, and N-representability barriers [1, 2, 3]. The computational evidence offered here is correspondingly weaker than in the main collection: it verifies the standard sub-lemmas the reasoning uses and exercises the mechanisms on small exactly diagonalizable proxies, and it is *not* asymptotic validation of any of the twelve structural laws. What each conjecture does provide is a crisp falsifier: an adverse scaling of one identified structural quantity refutes it. The value of the collection is that discipline, not a claim of proof.

Standard of evidence

Three separations from the main collection are stated once, here, and hold throughout.

- *Conditional, not exact.* Every polynomial-time statement below is conditioned on an explicit promise class. Where a step is not yet derived (polynomial conditioning, tractable separators, a certified polynomial-time structure for a nonconvex optimization, low approximation rank, and the like) it is stated as part of the promise rather than silently assumed. A promise class that quietly encodes its own conclusion would make the corresponding conjecture vacuous; the falsifiers are written to expose exactly that failure.

- *Sub-lemma verification only.* The accompanying script `verify/promise_lemmas.py` rederives the shared lemmas of Section 2 and three auxiliary bounds, and it runs small exact-diagonalization proxies. A passing check certifies an *ingredient* of the reasoning or a mechanism on a small instance; it is never evidence that one of the twelve structural laws holds asymptotically. The scripts are adapted, with attribution, from the source research package.

- *Known barriers respected.* Generic electronic structure is hard: the fermionic sign problem [1], N-representability is QMA-complete [2], and interacting-electron ground states are computationally intractable in the worst case [3]. Each conjecture is a structural escape hatch for a restricted class, not a challenge to these results.

1 Scope and the promise-polynomial solver

Fix a finite one-particle basis of size M and N electrons. The Hamiltonian H acts on the N-electron sector $\Lambda^N\mathbb{C}^M$ of dimension $\binom{M}{N}$. All twelve conjectures are first stated for this finite-basis problem; a continuum claim additionally requires a basis promise $M = \mathrm{poly}(N, 1/\varepsilon)$ with basis error at most $\varepsilon/3$, encouraged but not settled by the 2026 sparse-CI regularity analysis [4]. Coefficients, tolerances, and every promised conditioning or spectral constant are taken to have polynomial bit length, and every numerical primitive a conjecture invokes must be executable to the required accuracy in polynomial bit complexity; where this is not derived it is part of the promise.

Remark 1 (Promise-polynomial bounded-error solver)**.** *A method is* promise-polynomial *for a class if, for every instance in the class and every $\varepsilon > 0$, it returns an energy within ε of the ground state together with a computable certificate of that accuracy, in time polynomial in n, $1/\varepsilon$, and the promised constants. Every formal conjecture below asserts that its named construction is promise-polynomial on its promise class.*

2 Common lemmas

These are standard and are used repeatedly; each is re-derived numerically in the companion script. They are ingredients, not conclusions.

Lemma 1 (Rayleigh continuity)**.** *For normalized ψ_0, ϕ, $|\langle\phi, H\phi\rangle - \langle\psi_0, H\psi_0\rangle| \le 2\|H\|\,\|\phi - \psi_0\|$.*

Lemma 2 (Energy-to-fidelity under a gap)**.** *If $H\psi_0 = E_0\psi_0$ with spectral gap Δ and ϕ is normalized, then $\langle\phi, H\phi\rangle - E_0 \ge \Delta(1 - |\langle\psi_0, \phi\rangle|^2)$.*

Lemma 3 (Krylov row leverage)**.** *If U has orthonormal columns spanning a Krylov space K and $\ell_I = \|e_I^\top U\|^2$, then for every unit $x \in K$ and determinant subset S, $\|Q_S x\|^2 \le \sum_{I\notin S} \ell_I$.*

Lemma 4 (Natural-orbital freezing)**.** *With exact natural occupations and defect tail T, the probability of at least one frozen-orbital defect is $q \le T$; the projected normalized state obeys $\|\phi - \psi_0\| \le \sqrt{2T}$ and energy error $\le 2\sqrt{2}\,\|H\|\sqrt{T}$.*

Lemma 5 (Feshbach fixed-point perturbation)**.** *For $H = \left[\begin{smallmatrix} A & B \\ B^* & D \end{smallmatrix}\right]$ and $\Sigma(E) = B(D - E)^{-1}B^*$, if a branch $f(E) = \lambda(A - \Sigma(E))$ is an L-contraction and $\sup_E\|\Sigma - \Sigma_r\| \le \delta$, the fixed-point energy error is $\le \delta/(1 - L)$.*

Lemma 6 (CMI recovery)**.** *By Fawzi–Renner [13], a recovery channel has fidelity with $I(A : C \mid B) \ge -2\log F$, so the trace error of a single stitch is $O(\sqrt{I})$ and sequential errors accumulate by contractivity and the triangle inequality.*

Lemma 7 (Treewidth to polynomial)**.** *An algorithm of cost $\mathrm{poly}(n)\exp(O(t))$ is polynomial when $t = O(\log(n/\varepsilon))$ [11].*

3 The twelve conjectures

Conjecture 1 (Certified Influence-Graph Coupled Cluster)**.** *On the promise class below there are fixed $c, \alpha > 0$ and a polynomial-time procedure that, from a certified excitation-influence graph of logarithmic effective radius, returns a Full-CC energy within ε of the ground state together with a computable certificate of that accuracy.*

Promise class. Finite-basis, charge-neutral Hamiltonians in localized orbitals with $M = \mathrm{poly}(n)$, bounded orbital density and geometric degree; an isolated ground state with gap $\Delta \geq 1/\mathrm{poly}(n)$; a single-reference Full-CC root t^* at which the CC derivative $Df(t^*)$ is invertible with $\|Df(t^*)^{-1}\| \leq \mathrm{poly}(n)$; polynomial local Lipschitz constants transferring amplitude error to energy error; a certified influence representation with polynomially many seeds and bounded effective branching after locality aggregation; exponentially decaying weighted excitation-influence paths; and a polynomial-time computable aggregate upper envelope for the omitted-path tail, so certification never enumerates all omitted excitations.

Mechanism and first sub-lemma. Near a regular Full-CC root the analysis of Hassan–Maday–Wang [5] controls amplitude error by $\|t_R - t^*\| \leq 2\kappa_R \rho_R$, inverse-Jacobian norm times omitted residual; Lemma 1 then bounds the energy error. The conjectural ingredient is the exponential graph-tail law together with polynomial conditioning on the class.

Nearest literature. Selective coupled cluster and commutator screening [6], and residual-based Full-CC error control [5]; the combination with a computable omitted-path certificate and a promise-polynomiality claim is the candidate-new element.

Falsifier. A bounded-geometry gapped counterfamily whose outgoing influence $W_{\mathrm{out}}(R)$ decays no faster than inverse-polynomial in R, or with unavoidable super-polynomial Jacobian conditioning.

Conjecture 2 (Krylov-Leverage Closure Selected CI)**.** *On the promise class below there is $s = \mathrm{poly}(n, 1/\varepsilon)$ such that determinant selection by Krylov row-leverage followed by $O(\mathrm{polylog})$ residual-closure rounds yields a space V_S containing a normalized vector within $\varepsilon/(4\|H\|)$ of the ground state, in polynomial time.*

Promise class. A cheaply constructible reference ϕ with $|\langle \phi, \psi_0 \rangle| \geq 1/\mathrm{poly}(n)$, spectral width $\mathrm{poly}(n)$, and gap $\Delta \geq 1/\mathrm{poly}(n)$; for $q = \mathrm{polylog}(n/\varepsilon)$ the ground state lies within $\varepsilon/\mathrm{poly}(\|H\|)$ of $K_q = \mathrm{span}\{\phi, \ldots, H^{q-1}\phi\}$; the leverage distribution of an orthonormal basis of K_q has polynomial effective support after Hamiltonian-neighbor closure; and every compressed Krylov vector and above-threshold leverage/residual row is generable from a polynomial-size determinant frontier without scanning the full basis.

Mechanism and first sub-lemma. Lemma 3 makes the omitted leverage a rigorous uniform support bound for vectors inside K_q; the closure rounds are the conjectural ingredient.

Nearest literature. Heat-bath, CIPSI, fast-randomized-iteration, and time-evolved selected CI [7, 8, 9]; leverage-score selection from a Krylov space with residual closure and a two-term certificate appears new.

Falsifier. A promise-class family whose Krylov leverage has no polynomial effective support, or where closure fails to reach the target space in polylogarithmically many rounds.

Conjecture 3 (Multiscale Entanglement-Curvature Active Spaces)**.** *On the promise class below, selecting orbitals by entropy-curvature persistence across variational resolutions plus a logarithmic halo yields an active space of size $O(\mathrm{polylog}(n/\varepsilon))$ whose solver is polynomial and whose complementary correction has a computable polynomial-time bound completing the error to ε.*

Promise class. Systematically improving low-cost states $\psi^{(0)}, \ldots, \psi^{(K)}$ with computable one-orbital entropies $s_i^{(k)}$; statically correlated orbitals occur in $O(\mathrm{polylog}(n/\varepsilon))$ persistent multiscale curvature events while smooth dynamic correlation has bounded total curvature.

Mechanism and first sub-lemma. The score $c_i = \sum_k w_k \, |s_i^{(k+1)} - 2s_i^{(k)} + s_i^{(k-1)}|$ marks orbitals whose one-site entropy changes nonlinearly as resolution grows; Lemma 4 supplies the occupation safeguard for near-degenerate occupations. The polylogarithmic active-space bound is conjectural.

Nearest literature. Orbital-entropy active-space selection, `autoCAS` [10]; curvature/persistence of entropy across resolution as the selection invariant is the new element.

Falsifier. A gapped localized family in which an extensive set of orbitals has small but persistent curvature whose cumulative omission gives $O(1)$ energy error.

Conjecture 4 (Cumulant-Treewidth Reconstruction)**.** *On the promise class below, thresholding a two-body cumulant interaction graph to treewidth $O(\log(n/\varepsilon))$ while keeping the discarded-cumulant energy below budget yields a polynomial-time tree-decomposition solver for energy and RDMs within ε.*

Promise class. Edge weights $w_{ij} = \|V_{ij}\|_* \, \|\Lambda_{ij}\|$ from matched integral and connected-cumulant blocks; for every ε a threshold τ computable from certified interval-error RDMs whose discarded weight is $\leq \varepsilon/4$ and whose retained graph has treewidth $t = O(\log(n/\varepsilon))$. The cost of the certified interval bounds is part of the promise and may not be charged to a hidden oracle.

Mechanism and first sub-lemma. Lemma 7 turns logarithmic treewidth into polynomial contraction; the discarded-cumulant-energy budget controls the truncation.

Nearest literature. Orbital mutual-information graphs and tensor-network/treewidth methods; treewidth defined from thresholded 2-RDM cumulant blocks with an omitted-energy budget is the new element.

Falsifier. A counterfamily for which every edge-deletion set of cumulative dual-weight $\leq \varepsilon$ leaves treewidth $\Omega(n^a)$.

Conjecture 5 (Feshbach Rational External-Space Compression)**.** *On the promise class below, compressing the Q-space self-energy $\Sigma(E) = B(D - E)^{-1} B^*$ by a low-degree rational/Krylov approximant and solving the resulting fixed point returns the ground-state energy within ε in polynomial time.*

Promise class. P projects onto a polynomial reference space, $Q = I - P$; an interval $I \ni E_0$ with $\operatorname{dist}(I, \sigma(D)) \geq g \geq 1/\operatorname{poly}(n)$; a target branch of $A - \Sigma(E)$ isolated by $\geq 1/\operatorname{poly}(n)$; branch Lipschitz constant $L < 1 - 1/\operatorname{poly}(n)$; numerical coupling rank $k = \operatorname{poly}(n, \log(1/\varepsilon))$; and rational/Krylov approximability of the Q-resolvent to operator error δ using $r = \operatorname{polylog}(n/\varepsilon)$ applications of D, with every D-action, orthogonalization, and error estimate computable in polynomial time without materializing Q.

Mechanism and first sub-lemma. Lemma 5 converts a uniform resolvent approximation error δ into effective-Hamiltonian error $\delta/(1-L)$; the low rational degree and coupling rank are conjectural.

Nearest literature. Feshbach/downfolding and reduced-order Green's function models [12]; a certified low-degree rational compression of the CI self-energy with a contraction-map root bound is the new element.

Falsifier. A promised family where $\operatorname{dist}(E_0, \sigma(D))$ stays inverse-polynomial but the rational-approximation rank of Σ is exponential.

Conjecture 6 (Fermionic Conditional-Mutual-Information Stitching)**.** *On the promise class below, a parity-preserving recovery-map stitching of local patch solutions across A–B–C separators returns the ground state within ε in polynomial time.*

Promise class. Localized orbitals partitioned into A–B–C separators on a bounded-degree graph, in a parity-respecting fermionic representation; for every separator with buffer b, $I(A : C \mid B)_\psi \leq C \exp(-\alpha b)$ up to polynomial prefactors; and a patch/recovery representation of tractable size, $O(b)$ orbitals or width $O(b)$, with each patch solve and recovery map computable to inverse-polynomial precision in $\exp(O(b))\operatorname{poly}(n)$ time.

Mechanism and first sub-lemma. Lemma 6 bounds a single stitch error by $O(\sqrt{I})$ and accumulates them by contractivity; the constructive parity-preserving recovery on a tractable-separator class is the conjectural ingredient.

Nearest literature. Quantum Markov recovery [13], fermionic Gaussian recovery maps [14], and the 2026 universal-CMI-decay results [22]; the novelty is *not* CMI decay itself but the constructive electronic-structure stitching algorithm and its complexity theorem.

Falsifier. A gapped bounded-degree family with CMI bounded below by a constant across every logarithmic separator, or with small CMI but no polynomially constructible compatible parity-preserving recovery.

Conjecture 7 (Natural-Occupation Tail Certified Pruning)**.** *On the promise class below, retaining $O(\operatorname{polylog}(n/\varepsilon))$ correlation orbitals per bounded local domain drives the occupation-defect tail T below $(\varepsilon/\|H\|)^2$, certifying the frozen core/virtual to energy error ε in polynomial time.*

Promise class. Exact natural occupations n_p; for frozen sets F_0, F_1 the defect $T = \sum_{p \in F_0} n_p + \sum_{p \in F_1}(1 - n_p)$; approximate natural orbitals and certified occupation intervals computable in polynomial time; and the domain-summed T is $O((\varepsilon/\|H\|)^2)$ after retaining $m = O(\mathrm{polylog}(n/\varepsilon))$ correlation orbitals per bounded local domain.

Mechanism and first sub-lemma. Lemma 4 gives $q \leq T$ and the energy bound $2\sqrt{2}\,\|H\|\sqrt{T}$; the decay of T under polylogarithmic retention is the conjectural promise.

Nearest literature. Natural orbitals, occupation truncation, and PNOs [15]; the explicit global freezing-error certificate $q \leq T$ with a promise-class decay stopping rule is the new element.

Falsifier. A bounded-density gapped family for which every polynomial local-domain retention leaves $T = \Omega(1)$ at fixed accuracy.

Conjecture 8 (Integral-Frustration Local-Orbital Sign-Gap)**.** *On the promise class below there are fixed c, C, α with $\Delta_s(U) \leq Cn^c F_k(U) + Cn^c \exp(-\alpha k)$, and a polynomial algorithm finds block-local U with $F_k(U) \leq \varepsilon/\operatorname{poly}(n)$ at $k = O(\log(n/\varepsilon))$, in whose basis a specified projector-QMC scheme has at worst polynomial sign penalty for the projection time reaching energy ε.*

Promise class. $\Delta_s(U) = E_0(H(U)) - E_0(H_{\mathrm{abs}}(U)) \geq 0$ for the sign-relaxed Hamiltonian; U a product of rotations on $O(1)$-diameter blocks; bounded-degree integral-generated interaction graph after certified thresholding of an energy-negligible integral tail; $F_k(U)$ a polynomial-time weighted sum of frustrated signed loops of length $\leq k$ in the *integral-generated* graph, not the exponential determinant graph; and a certified polynomial-time structure for the continuous block-rotation optimization (a separable/convex factorization or another verifiable oracle), since polynomial parameter count alone does not make a nonconvex optimization tractable.

Mechanism. The conjecture bridges an integral-level frustration certificate to the determinant sign gap and then to projector-QMC variance; the optimization tractability is stated as an explicit promise, not assumed from parameter count.

Nearest literature. Basis rotations for sign mitigation [16, 17, 18]; the integral-level loop-frustration upper certificate with a scaling theorem is candidate-new, and this is the highest collision-risk item in the collection.

Falsifier. A promised family and rotations with $F_k(U_n) \to 0$ for all $k = O(\log n)$ but $\Delta_s(U_n) \geq c > 0$, or average sign still $\exp(-\Omega(n))$.

Conjecture 9 (Möbius Connected-Fragment Exponential Decay)**.** *On the promise class below, the Möbius connected increment $\Delta(S)$ of embedded correlated fragment energies decays exponentially in fragment diameter after subtraction of classical electrostatics and mean-field polarization, yielding a computable tail and a polynomial-time bounded-degree connected-cluster summation to energy ε.*

Promise class. A bounded-degree graph of localized fragments; a size-consistent embedded correlated energy $E_{\mathrm{corr}}(S)$; the connected increment $\Delta(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} E_{\mathrm{corr}}(T)$ with the connected-cluster adaptation; neutrality, a gap, bounded density, $O(1)$ spin-orbitals per fragment, and explicit subtraction of classical electrostatics/mean-field polarization.

Mechanism. Second finite differences of open-cluster ground energies are a linked-cluster/Möbius proxy; the conjectural ingredient is the uniform exponential bound on embedded increments, whose summation is controlled by connected bounded-degree cluster counting rather than unrestricted n^k enumeration.

Nearest literature. The method of increments and local-correlation expansions [19]; a uniform exponential bound on embedded Möbius increments yielding a promise FPTAS is the new element.

Falsifier. A neutral gapped bounded-degree family with $|\Delta(S)|$ having only algebraic non-summable tails after the stated subtraction.

Conjecture 10 (Globally Budgeted Hierarchical Pair Natural Orbitals)**.** *On the promise class below, casting all pair ranks as one certified global resource-allocation problem yields, in polynomial time, a PNO truncation meeting a prescribed total discarded-weight budget with total energy error ε, with strict savings over uniform thresholds on heterogeneous instances.*

Promise class. Ordered pair-correlation singular values $\sigma_{a,j}$ with tail $\tau_a(r) = \sum_{j>r} \sigma_{a,j}^2$; a computable local sensitivity bound L_a converting pair error into total energy error; $O(n)$ strong pairs after a distance hierarchy; polylogarithmic ranks for each target tail; polynomially bounded sensitivities; certified tail/sensitivity intervals; and a separable diminishing-returns (discrete-convex) rank-cost structure making the global allocation polynomially solvable.

Mechanism. Global sorting of pair modes against a total budget replaces per-pair uniform thresholds; the discrete-convex allocation promise makes the budgeted optimization polynomial.

Nearest literature. PNO thresholds and complete-PNO extrapolation [20]; the single certified global sensitivity-weighted allocation with a complexity statement is the new element.

Falsifier. A family where every computable polynomial L_a must be exponentially loose, or where cross-pair interference invalidates any summable tail bound of this form.

Conjecture 11 (Multipole-Separated Cumulant-Order Bound)**.** *On the promise class below, a single additive remainder inequality couples far-field multipole order p and connected-cumulant order k, so that treating separations $\leq R$ exactly, far cells by multipoles through order p, and the correlated solver through cumulant order k achieves energy error ε in polynomial time.*

Promise class. Neutral, gapped, bounded-density systems in $O(1)$-orbital cells of diameter a on a bounded-degree near-field graph; bounded response moments; and exponential clustering of connected *quantum* correlations after separating classical electrostatics.

Mechanism. The near-field low-rank far-field split is standard; the conjectural ingredient is the joint remainder bounding neglected higher connected cumulants after multipole separation, summed by bounded-degree connected-cluster counting.

Nearest literature. Fast-multipole/low-rank Coulomb methods and local-correlation expansions, separately; the joint far-field/cumulant remainder inequality is the new element.

Falsifier. A promised family where, after classical-electrostatic separation, connected quantum correlations do not cluster exponentially, or the joint remainder is not additively controllable.

Conjecture 12 (Chordal Local N-Representability Dual Certificate)**.** *On the promise class below, increasing chordal bag width until the certified bracket closes gives a polynomial-time algorithm returning $E_0 \in [L_b, U_b]$ with $U_b - L_b \le \varepsilon$, where L_b is the optimum of a principal-block N-representability SDP relaxation and U_b any variational upper bound.*

Promise class. A weighted orbital cumulant graph with a chordal cover of bags B_j; global 1-/2-RDM variables (or overlap-consistent bag copies) subject only to constraints every global N-electron 2-RDM necessarily satisfies (principal-submatrix $D/Q/G$ positivity per bag, globally valid trace/contraction/spin identities restricted consistently, optionally principal T_2 blocks); *no* fixed local particle number; every Hamiltonian objective coefficient represented by a constrained entry or handled by an explicit rigorous residual bound; treewidth $b = O(\log(n/\varepsilon))$; and exponential local-hierarchy completeness, $U_b - L_b \le C n^c \exp(-\alpha b)$.

Mechanism. The bracket is rigorous unconditionally: every exact global 2-RDM restricts to feasible principal $D/Q/G$ blocks and satisfies the shared identities, so L_b is a genuine lower bound [21], while U_b is variational; with $b = O(\log(n/\varepsilon))$ the per-bag SDP has polynomial dimension. The single conjectural ingredient is the exponentially closing gap on the promise class.

Nearest literature. Variational 2-RDM $D/Q/G$ lower bounds [21], chordal PSD completion, and fermionic RDM matrix completion [23, 24]; the low-treewidth principal-block relaxation paired with a variational upper hierarchy and a conjectured exponentially closing certified bracket is the new element.

Falsifier. A promised low-treewidth gapped family with $U_b - L_b \ge c > 0$ for all $b = O(\log n)$, from a genuinely global N-representability obstruction.

4 Sub-lemma audit and sanity checks

The companion script re-derives Lemmas 1–7 and three auxiliary bounds used by Conjectures 5, 8, and 12 (block spectral lower bounds; determinant sign-gap nonnegativity; and the block-Gershgorin lower-bound proxy for the N-representability SDP), and it exercises each mechanism on a small exactly diagonalizable instance. Every re-derived inequality holds to machine precision, and each proxy reproduces the qualitative mechanism it targets. This certifies ingredients and mechanisms only; it is not asymptotic evidence for any of the twelve structural laws, and no small-instance "mechanism supported" label should be read as such.

Relation to the main collection

This companion is deliberately separated from *Three hundred nineteen conjectures in mathematics*. That collection states exact, unconditional laws, each reproduced numerically to the stated precision; this one states conditional promise-class algorithmic laws whose actual truth is not numerically validated here. The two are kept apart precisely so that the main collection's "every claim verified" standard is never diluted, and so that these conditional conjectures are judged on their own terms: a crisp promise class and a single falsifiable structural quantity apiece.

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